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Vehical Tracking and Speed Estimation of Moving Vehicles for Traffic Surveillance Applications

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Abstract- Now-a-days traffic in two-tier cities has increased lot, so surveillance system is needed to monitor the traffic and avoid unwanted delays and accidents. Vehicles speed estimation can be done by analyzing the video captured. The conventional or today's method used to control or measure the speed of the vehicles is through using RADAR device and other models like constant g-factor, sequence method and motion median. The limitations of this device or models are accuracy is bit less and costly and does not capture long distance images. Different techniques have been used for speed estimation; here in this paper we propose a mixture of Gaussian technique to estimate the speed. In speed estimation we need to detect vehicle from the captured video and analyze the video considering the centroid at each instance of a frame and distance will be calculated. The proposed method consists of mainly four phases background subtraction using Mixture of Gaussian, feature extraction using Shi-Tomasi technique, vehicle tracking, distance and speed calculation. The effectiveness of using Gaussian mixture model was evaluated with a sample video data set and the actual speed and the determined speed comparison is shown in the below results. The vehicle speed is estimated using distance travelled by the moving vehicle over number of frames and frame rate.

Keywords— Background subtraction, Vehicle Detection, Speed Estimation, GMM

I. INTRODUCTION

Monitoring traffic has become an important task in urban areas, so image processing and computer vision techniques are used to monitor manage and to determine speed of the moving vehicles. These techniques are more efficient compared to present and other techniques in terms of accuracy and processing speed. As the population has increased the vehicles on road is increasing tremendously and hence traffic surveillance systems based on capturing video and processing it to determine the speed has become an important issue to control the accidents and traffic jam problems. The traffic surveillance systems are very useful to monitor and manage various traffic conditions such as traffic management, prevention of accidents and secure transportation. Hence, determining the speed of vehicle has become a challenging task. Traditionally RADAR systems were used for monitoring the speed of vehicles. The major drawback associated with RADAR is its cost and accuracy, the device is quite expensive and is less accurate and the biggest drawback is that a line of sight connection needed between RADAR and the vehicles. So to overcome the limitations in existing methods, various image processing techniques are used. The Gaussian mixture

model is one among those techniques which is efficiently used for vehicle detection and vehicle speed estimation.

In urban areas the traffic monitoring cameras are stationary. They will be located above the ground level to get a clear view of vehicles moving on road. The vehicle detection from stationary camera will be much easier as compared to cameras which capture the dynamic change of the vehicles and its surrounding environment. In this paper we are analyzing the videos captured by stationary cameras, and to detect multiple vehicles optical-flow based technique is used. Using this technique multiple images are recognized at different times, optical-flow based technique indirectly detects obstacles by analyzing the velocity field.

Object detection techniques can be classified depending on the background subtraction and frame differencing.

i. Considering the Threshold value:

Here we are going to consider the intensity level of an object eliminating the background scenes. By considering threshold intensities for a region we will be able to separate vehicle from background. So choosing the threshold value becomes an important task. If the threshold value is less then we may not detect the exact vehicle. And if the threshold value is more, then chances are there to detect the noise along with the vehicle. So deciding the threshold intensities is very difficult using this approach. So gray-scale morphological operations have been used for object detection.

ii. Gaussian mixture background model:

The best approach for object detection is through using background subtraction. Different techniques are available but the most efficient one is Gaussian mixture Model (GMM). The GMM basically operates at pixel level, and are most widely used for still background. In the background subtraction method we detect the foreground objects as the difference between the two consecutive frames. By using the GMM model for background subtraction, frame pixels are removed from the required video to obtain the desired results. It should react to changes of frames due to start and stopping of the object.

II. RELATED WORK

A. Vehicle Detection:

A number of techniques are available for vehicle detection from a captured video. Detection of moving vehicle accurately

from the video is a challenging task. The moving vehicle detection provides a classification of the pixels in the video frames either for foreground or background data. The approaches for vehicle detection are Frame differencing, background subtraction algorithm [1]. Frame differencing method can be used to eliminate the still objects i.e. which are not moving, and will consider the objects which are under motion. The main drawback of this method is slow moving objects are not detected, so improvisation is needed for frame differencing technique to detect slow motion objects too.

Background subtraction means that the background or the static scene is extracted from the video image. Because the camera is fixed, each pixel in the image has a corresponding value which is basically fixed over a period of time. The purpose of background subtraction is to find the background value of each point of the image. We have used Mixture of Gaussian technique to detect the moving vehicle since it overcomes the limitations of the other methods. The best approach for removing the background and considering the vehicle from the video is through background subtraction where each frame of the video is compared with reference or a background model, and the pixels where deviation is observed from the background are considered to be the vehicles.

Once the moving vehicle is detected from the image frames, then the next step is to track the vehicle from the video sequence, by tracking the vehicle we will be able to determine the velocity, acceleration and position of vehicle at different instance of time interval. In traffic surveillance or other applications it is desired to keep track of velocity and speed of the vehicles.

B. Feature Extraction:

J. Shi and C. Tomasi made a small modification to the Harris corner detection technique which shows better results compared to Harris Corner Detector. Shi- Tomasi algorithm is advanced version of Harris corner detection. However, it is much better than the original corner detector [5].

For Shi-Tomasi, it's calculated like this: $R = \min(\lambda_1, \lambda_2)$. If both λ_1 and λ_2 are greater than a certain value, then this region is "accepted" as corners. If either λ_1 or λ_2 is less then it is not a corner.

C. Vehicle Tracking:

Tracking of vehicles from a video where vehicles are in motion is a challenging task. Many difficulties may arise in vehicle tracking such as no rigid vehicle structure, changes in appearance pattern and occlusions between the object-to-scene. In general, tracking can be defined as the problem of determining the trajectory motion of an object as it moves around in a scene. Here the object tracker assigns a label to all the tracked vehicles. In other words; a tracker can also provide centric-information, orientation and shape of a vehicle.

The KLT algorithm [4] is used to track the features in consecutive frames. This algorithm takes two frames as inputs and compares their features depending upon certain criteria.

The features in each frame are detected and the good features are selected by Shi- Tomasi algorithm. The feature selection depends on the minimum distance between the features, and the Eigen value. Object tracking using Optical Flow [5] refers to the process of tracing the moving object in progression of frames. The task of tracking vehicle is performed by feature extraction in a frame and discovering the objects in sequence of frames. By using the corner values of each object in each frame, we can determine the position and velocity of the moving object.

III. PROPOSED METHOD

In the proposed method, the speed is estimated for the vehicle which is coming towards camera by tracking its motion of through sequence of images. Here the proposed system mainly consists of 5 components as shown in fig.1. In preprocessing the captured video is converted into frames and the noise is removed using Median Filter technique. In moving vehicle detection, background Subtraction (Mixture of Gaussian) method is used. The foreground thus detected is cleaned by applying morphological operations such as opening. Connected Component Method is used to detect each object and. In feature extraction corners are detected using Shi-Tomasi technique. In vehicle tracking, vehicles are tracked using optical flow technique. In speed estimation, the centroids of each vehicle are calculated. Centroids are tracked over multiple frames. Velocity is calculated using distance traveled by vehicle and time taken. Figure. 1 shows Block Diagram for proposed method

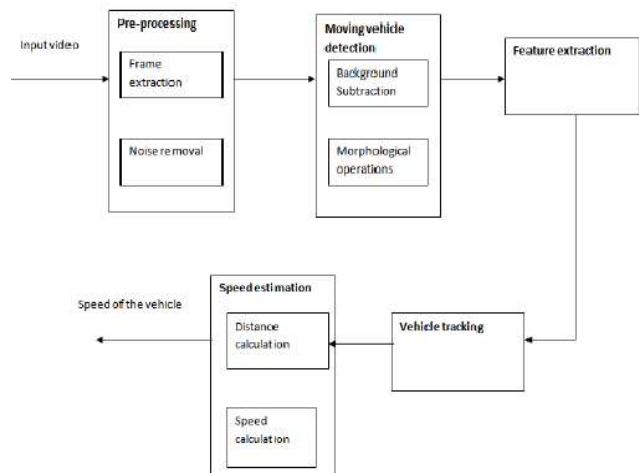


Figure.1 Block diagram of proposed system

The proposed system model diagram is shown in figure 1. The proposed system can be divided mainly into 5 modules preprocessing, vehicle detection, Feature extraction, vehicle tracking, and speed estimation:

A. Preprocessing:

Video preprocessing is the first stage which involves frame extraction and video enhancement. In preprocessing the given

video is converted into n number of frames and frame rate are extracted. In the next stage, quality of video is improved by reducing the noise from background. And for this, median filter technique is used. This is a nonlinear filtering technique, often used to remove noise from background and hence obtain an enhanced foreground. Such technique is often a pre-processing step to improve the accuracy for later processing. Although there are various methods we prefer median filter since it preserves the edges while removing noise. The result of preprocessing is as shown in figure.2



Figure.2 After Noise Removal

B. Moving Vehicle Detection:

The next step is background subtraction. It also known as Foreground Detection, it is a technique in the field of image processing where the foreground of image is extracted. The image's regions of interest are vehicles in its foreground. Object localization often makes use of this technique. Here it is assumed that every pixel's intensity values in the video can be modeled using a Gaussian mixture model (MOG). A general rule determines which intensities are of the background and that of foreground. The pixels that do not match to these are known as foreground pixels. Foreground pixels are grouped using 8 connectivity connected component analysis. The result of background subtraction is as shown in figure.3

The MOG algorithm:

- It chooses the component (the Gaussian) randomly with probability $P(\omega_i)$
- It samples a point $N(\mu_i, \sigma^2 I)$.
Let's suppose to have: $x_1, x_2, \dots, x_N$
- $P(\omega_1), \dots, P(\omega_K), \sigma$

We can obtain the likelihood of the sample as follows:

$$P(x | \omega_i, \mu_1, \mu_2, \dots, \mu_K)$$

What we really want to maximise is $P(x | \mu_1, \mu_2, \dots, \mu_K)$ (probability of a datum given the centres of the Gaussians).

$$P(x | \mu_i) = \sum_i P(\omega_i) P(x | \omega_i, \mu_1, \mu_2, \dots, \mu_K)$$

The base to write the likelihood function:

$$P(\text{data} | \mu_i) = \prod_{i=1}^N \sum_i P(\omega_i) P(x | \omega_i, \mu_1, \mu_2, \dots, \mu_K)$$

We should maximise the likelihood function by calculating

$$\frac{\partial L}{\partial \mu_i} = 0$$

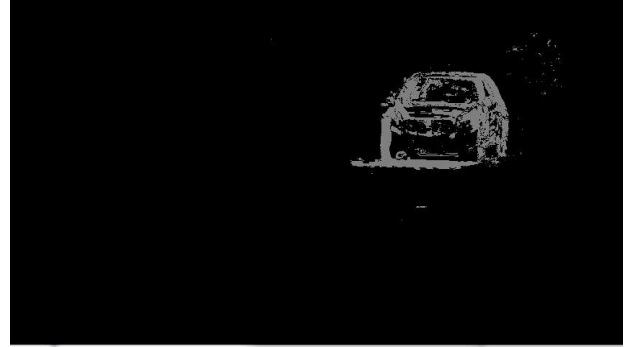


Figure. 3. After background subtraction

Once the foreground is extracted from background it has to be cleaned. The detected foreground is subjected to cleaning using a set of morphological operations such as opening. If erosion is followed by dilation, then the operations is called as opening. It is used to remove small objects from the foreground of an image and place them in the background. The result of morphological operation is as shown in figure.4



Figure. 4. After Morphological operation

C. Feature Extraction:

Corner detection is a widely used approach to extract certain kinds of features. It is mainly used to deduce the contents of an image. The Harris corner detector has some corner selection criteria. A score is calculated for each pixel in the image, and if the score is above a threshold value, then the pixel is marked as a corner otherwise it is not.

J. Shi and C. Tomasi made a small modification to the Harris corner detection technique which shows more accuracy compared to Harris Corner Detector.

The scoring function in Harris Corner Detector was given by:

$$R = \lambda_1 \lambda_2 - k (\lambda_1 + \lambda_2)^2$$

Instead of this, Shi-Tomasi proposed: $R = \min(\lambda_1, \lambda_2)$. If R value is greater than a threshold value, it is considered as a corner. Shi-Tomasi gives more accuracy than Harris corner detection method. The result of feature extraction is as shown in figure.5



Figure 5. Corner detection of cars

D. Vehicle Tracking

Vehicle tracking refers to the process of tracing the moving object in progression with respect to frames. The task of tracking is performed by feature extraction (Shi-tomasi method) of objects in a frame and discovering the objects in sequence. Considering the location values of object in each frame, we determine the position of the moving object. Optical flow is the pattern of motion of image objects between two consecutive frames caused by the movement of camera or object. It is 2D vector field and each vector is a displacement vector showing the movement of points from first frame to the next. The result of vehicle tracking is as shown in figure.6

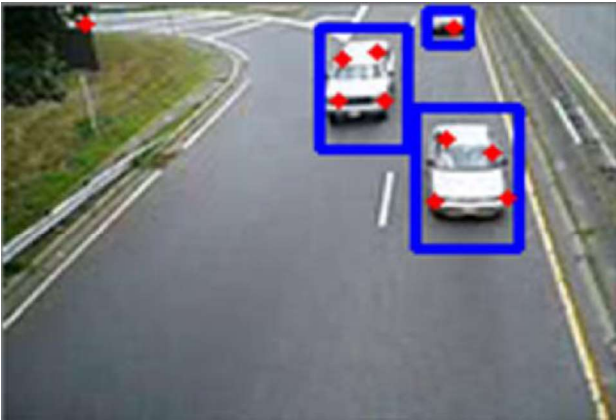


Figure 6. Multiple vehicle tracking

E. Speed Estimation:

Distance is calculated using the centroids of previous and next frames. For distance calculation, Euclidean distance is used. Time is calculated as vehicle enters the ROI. Initially the speed is calculated using the formula
If (x_1, y_1) is centroid of vehicle in first frame and (x_2, y_2) is centroid of second frame then
Distance travelled = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Speed = Distance travelled / Time taken.



Figure 7. Instantaneous Speed of single vehicle

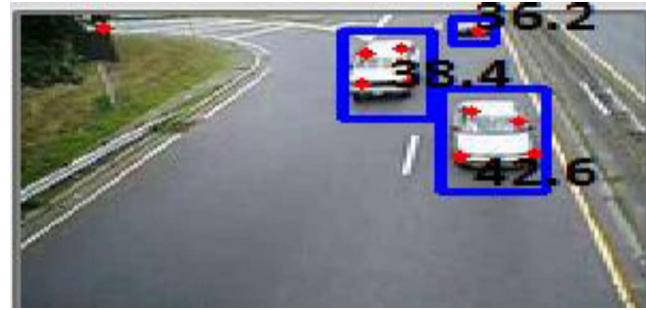


Figure 8 Instantaneous Speed of multiple vehicles

IV. RESULTS AND DISCUSSIONS

The table 1 shows the experimental results for speed estimation of different cars.

Table 1: Speed Evaluation

Sl.no.	Distance covered(m)	Actual speed(km/hr)	Estimated speed(km/hr)	Error rate
Car1	25	43	42.6	-0.4
Car2	21	39	39.4	+0.4
Car3	18	36	36.2	+0.2

From the above table we can infer that results have minimum error rate which can be further improved.

Comparative Study of Corner Detection Techniques:

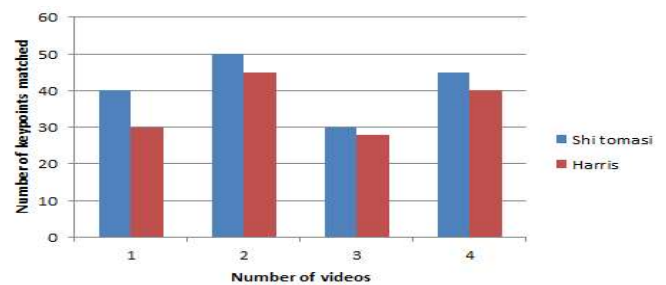


Figure 8. Comparison between Shi-Tomasi and Harris corner detection techniques

The above graph shows that Shi-tomasi technique for corner detection gives more accurate results as compared to Harris corner detection. So, Shi tomasi technique is preferred.

V. CONCLUSION

Vehicle speed estimation is often useful in system and other image processing applications. Initially the quality of the video is improved by applying noise removal technique such as Median filter on the input video for better results. Next phase is to subtract the background or the static scene and obtain only the foreground. Here there are many techniques and the one we have used is- Mixture Of Gaussian. The foreground thus obtained is cleaned by applying morphological techniques such as opening. In the next phase, vehicle tracking is done using Optical flow. And the centroids of all objects at each instance is calculated to make the prediction of its future motion. Finally the speed is estimated for moving vehicles using distance covered by the vehicle and time calculated.

REFERENCES

- [1] Stefan Leutenegger, Margarita Chli and Roland Y. Siegwart. "BRISK: Binary Robust Invariant Scalable Keypoints". In *Mechatronics and Automation (ICMA)*, 2014 IEEE International Conference on, pages 553–558. IEEE, 2014.
- [2] Jianbo Shi and Carlo Tomasi. "Good features to track". In *Computer Vision and Pattern Recognition, 1994. Proceedings CVPR'94.*, 1994 IEEE Computer Society Conference on, pages 593–600. IEEE, 1994.
- [3] Vamsi Krishna Madasu and Madasu Hanmandlu. "Estimation of vehicle speed by motion tracking on image sequences". pages 185–190, 2010.
- [4] Manisha Chaple and SS Paygude. "Vehicle detection and tracking from video frame sequence." *International Journal of Scientific & Engineering Research*, 4(3):1–7, 2013.
- [5] Abbas Dehghani and Pourmohammad, "Single Camera Vehical Speed Measurement". *IEEE Machine Vision and Image Processing (MVIP)*, 2013 8th Iranian Conference pages 190–193.
- [6] A. Nurhadiyatna; B. Hardjono; A. Wibisono; I. Sina; W. Jatmiko; M. Anwar Ma'sum; P. Mursanto "Improve Vehicle speed estimation using Gaussian mixture model And hole filling algorithm "Advanced Computer Science and Information Systems (ICACSIS), 2013 International Conference on year 2013 IEEE publications.